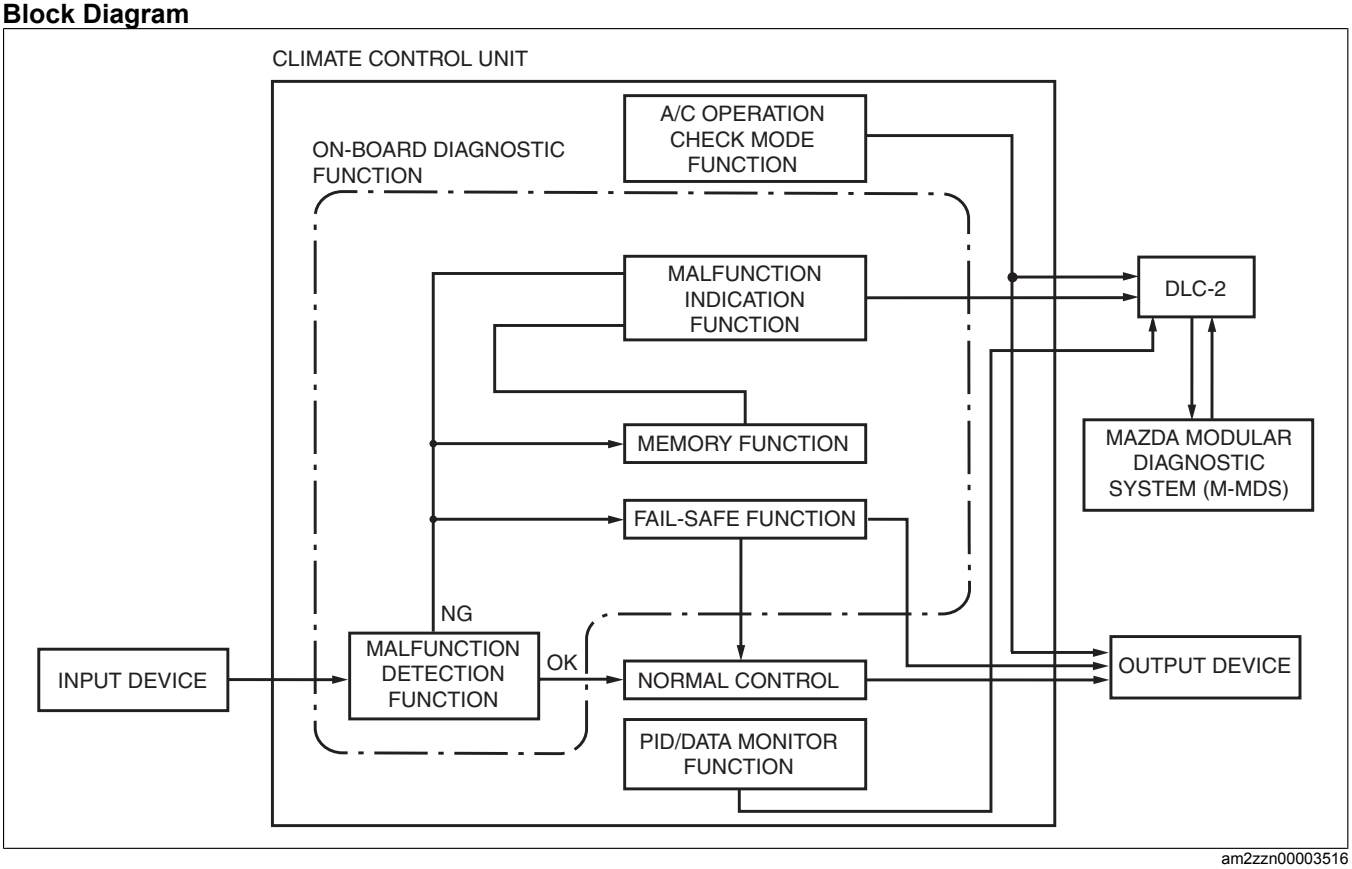
ON-BOARD DIAGNOSTIC

id070200126600

Outline

- The on-board diagnostic system has on-board diagnostic, PID/data monitor, and A/C operation check functions.
- The on-board diagnostic function has the following functions:
  - — Malfunction detection function
    - Detects errors in input and output signals
  - — Memory function
    - Stores detected malfunctions
  - — Malfunction indication function
    - Displays detected malfunctions
- The malfunction indication function, PID/data monitor, and A/C operation check functions can be performed by connecting the Mazda Modular Diagnostic System (M-MDS) to the DLC-2.

Block Diagram



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Function

Malfunction detection function

- The malfunction detection function detects errors in the input and output signals. (Ignition is switched ON (engine off or on))
- If a malfunction is detected, a DTC is output to the DLC-2 through the malfunction indication function.
- At the same time, malfunction detection results are sent to the fail-safe and memory functions.

Memory function

- The memory function stores malfunctioning signal systems which are detected to be malfunctioning by the malfunction detection function.
- The stored malfunction information is not cleared even when the ignition is switched off or the malfunction is repaired.
- The stored malfunction information can be cleared by connecting the Mazda Modular Diagnostic System (M-MDS) to the DLC-2 and operating it.

Malfunction indication function

- The malfunction indication function outputs present or past malfunctions to the DLC-2 as DTCs.
- The DTCs can be read by connecting the Mazda Modular Diagnostic System (M-MDS) to the DLC-2 and operating it.

Malfunction display mode

- The climate control unit detects present and past malfunctions in the control system circuits (open/short circuits) and indicates the DTCs shown in the table on the Mazda Modular Diagnostic System (M-MDS).
- Clear past malfunctions after completing repairs because once a past malfunction is stored, it will remain stored even after the malfunction has been repaired.
- The stored past malfunction can be cleared by connecting the Mazda Modular Diagnostic System (M-MDS) to the DLC-2 and operating it.

DTC table

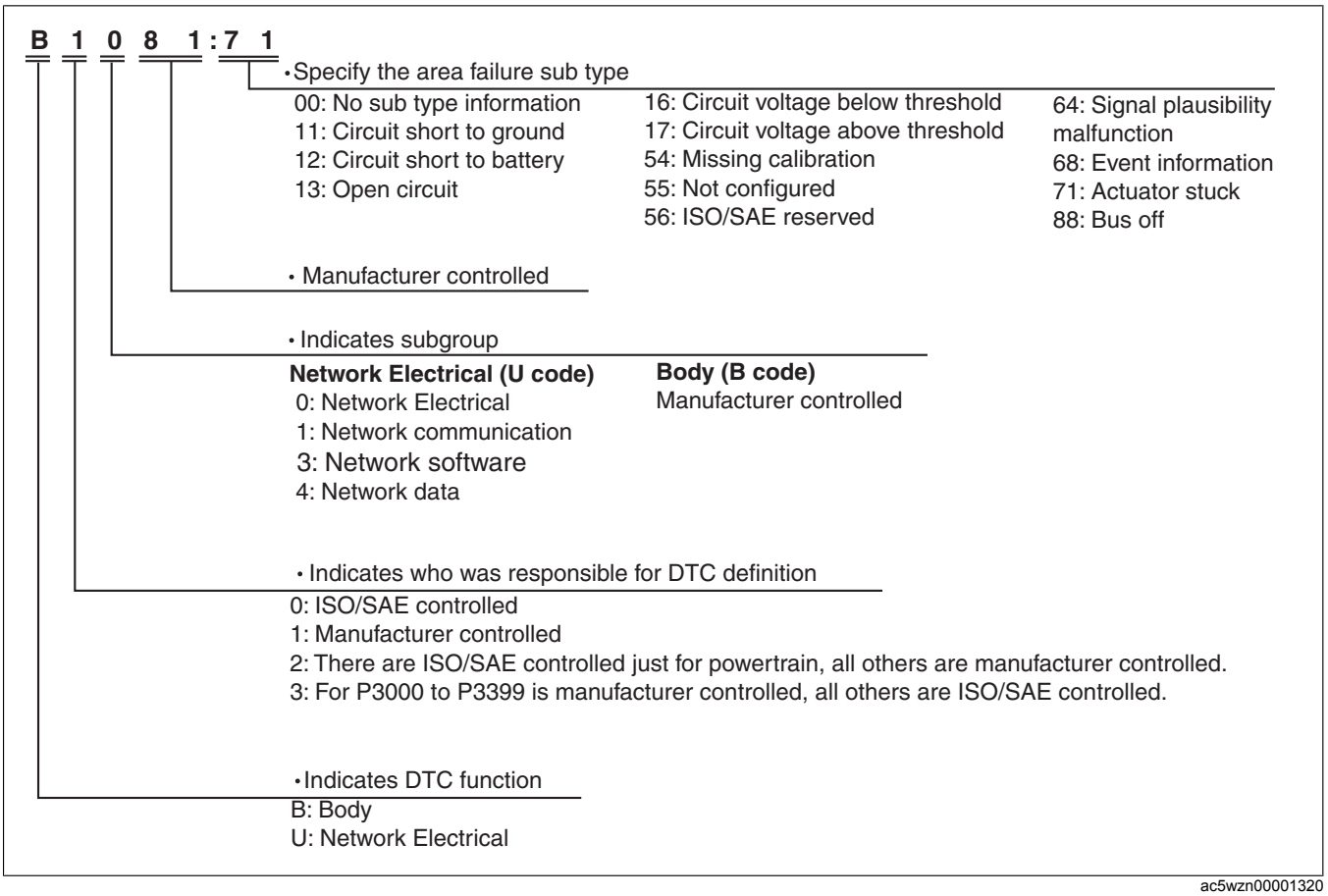
X: Applicable
—: Not applicable

DTC No.	Warning/ indicator light	System malfunction location	Fail-safe function	Drive cycle	Self test type*1	Memory function
B1086:71	—	Airflow mode actuator motor lock	X	—	C	X
B10BE:12	—	Solar radiation sensor circuit short to power supply	X	—	C, D	X
B10BE:13	—	Solar radiation sensor circuit open	X	—	D	X
B13AA:71	—	Air mix actuator motor lock	X	—	C	X
B13AB:12	—	Air mix actuator (potentiometer) circuit short to power supply	X	—	C, D	X
B13AB:13	—	Air mix actuator (potentiometer) circuit short to power supply	X	—	C, D	X
B1A61:11	—	Cabin temperature sensor circuit short to ground	X	—	C, D	X
B1A61:13	—	Cabin temperature sensor circuit open	X	—	C, D	X
B1B71:11	—	Evaporator temperature sensor circuit short to ground	X	—	C, D	X
B1B71:13	—	Evaporator temperature sensor circuit open	X	—	C, D	X
B1C1C:12	—	Airflow mode actuator (potentiometer) circuit short to power supply	X	—	C, D	X
B1C1C:13	—	Airflow mode actuator (potentiometer) circuit open	X	—	C, D	X
U0010:88	—	Unit communication error (MS-CAN)	—	—	C	X
U0155:00	—	Communication error with instrument cluster	—	—	C, D	X
U0423:68	—	Invalid data received from Instrument cluster	—	—	C, D	X
U200D:11	—	Climate control unit circuit voltage (**+5V**) circuit short to ground	—	—	C, D	X
U2300:54	—	Configuration error (data not received)	X	—	C, D	X
U2300:55	—	Configuration error (not configured)	X	—	C, D	—
U2300:56	—	Configuration error (ineffective/non-interchangeable data read)	X	—	C, D	X
U2300:64	—	Configuration error (error value read)	X	—	C, D	X
U3003:16	—	Climate control unit power supply circuit voltage below threshold (**10 V or less**)	—	—	C, D	X
U3003:17	—	Climate control unit power supply circuit voltage above threshold (**17.3 V or more**)	—	—	C, D	X
U300E:13	—	Climate control unit power supply circuit open (IG1)	—	—	C, D	X

*1 : C: CMDTC self test, D: ODDTC self test

DTC 7-digit code definition

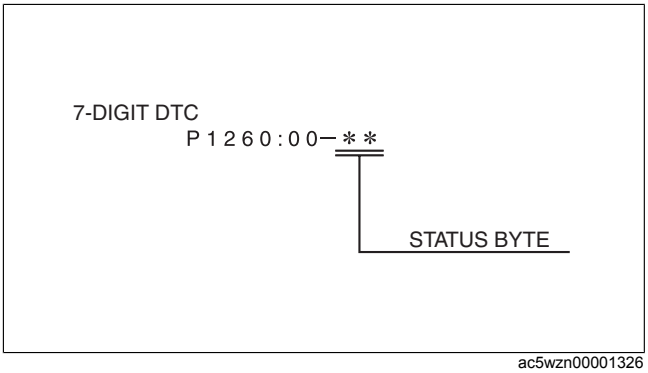
- When related systems or components have failed, the CM stores the DTC of the malfunctioning part in the CM memory, and allows for the retrieval of the stored data using the scanning tool when necessary. The DTCs are indicated using seven digits. Each digit indicates the following.



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Status byte for DTC

- The two digits (after hyphen) after the 7-digit DTC.
- A code which indicates the pending code, current/ past malfunction status, or warning illumination status.
- Can be read by performing a CMDTC self-test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the Mazda Modular Diagnostic System (M-MDS) when reading the DTC.



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PID/data monitor

- The PID/data monitor function is used for optionally selecting input signal monitor items preset in the climate control unit and reading them out in real-time.

PID/data monitor table

PID	Unit/Condition	Data contents	Climate control unit terminal
A/C_SW	Off/On	• Off: A/C switch is off. • On: A/C switch is on.	—
B_MT_RLY_CS	Off/On	• Off: Blower relay is off. • On: Blower relay is on.	1S, 1U (CAN)
ENG_C_TMP	°C, °F	Engine coolant temperature is displayed.	1S, 1U (CAN)
EVA_TMP_SEN	°C, °F	Evaporator temperature is displayed.	1L
F_REC_CS	Off/On	• Off: Forced recirculate control is off. • On: Forced recirculate control is on.	—
INC_TMP_SEN	°C, °F	Cabin temperature is displayed.	1H
OUT_CAR_TMP	°C, °F	Ambient temperature is displayed.	—
R/DEF_CS	Off/On	• Off: Rear window defroster is turned off. • On: Rear window defroster is turned on.	—

PID	Unit/Condition	Data contents	Climate control unit terminal
R/DEF_SW	Off/On	• Off: Rear window defroster switch is off. • On: Rear window defroster switch is on.	—
REC_SW	Off/On	• Off: RECIRCULATE switch is off. • On: RECIRCULATE switch is on.	—
SLR_R_SEN	W/m^2	Solar radiation amount is displayed.	K
STOP_ST	Available/ Not Available/ Error	• Available: i-stop permit request signal is sent. • Not Available: i-stop inhibit request is sent. • Error: i-stop signal error is sent.	K

A/C Operation Check Mode

- The A/C operation check mode forcibly operates the output devices as shown in the operation check table regardless of the signals input to the climate control unit.
- During an A/C operation check mode operation, indication in the display panel and the illumination of the indicator light on the switches are automatically controlled.
- Determine the malfunctioning part by visual inspection, listening to the operation sound, placing a hand on the air vents, or checking that each transition is as indicated in the operation check table.

Mazda Modular Diagnostic System (M-MDS) display	Target part	Reference
Air Mix Actuator	Air mix actuator Air mix door	(See Air mix actuator.)
Air conditioning compressor	A/C compressor	(See A/C compressor.)
Air Intake Actuator	Air intake actuator Air intake door	(See Air intake actuator.)
Blower Motor Speed	Blower motor	(See Blower motor.)
Airflow Mode Actuator	Airflow mode actuator Airflow mode door	(See Airflow mode actuator.)
Illumination Of All Indicator Lights	Climate control unit	(See Indicator light.)

Air mix actuator

Operation

Step	Air mix actuator	Airflow mode actuator	Blower speed	Magnet clutch	Air intake actuator	Rear window defroste
1	**0 %**	VENT	5th	ON	FRESH	OFF
2	**100 %**					

Indicator light

Step	Recirculate switch indicator light	Rear window defroster switch indicator light	A/C switch indicator light
1	Not illuminated	Not illuminated	Illuminated
2			

Note

- After **approx. 4 s** have elapsed, the operation step moves to the next step.
- When the last step is finished, the operation is repeated from Step 1.
- To prevent super-cooling, the magnetic clutch turns off despite the A/C forced override.

A/C compressor

Operation

Step	Air mix actuator	Airflow mode actuator	Blower speed	Magnet clutch	Air intake actuator	Rear window defroster
1	**0 %**	VENT	5th	ON	FRESH	OFF
2	**100 %**			OFF		

Indicator light

Step	Recirculate switch indicator light	Rear window defroster switch indicator light	A/C switch indicator light
1	Not illuminated	Not illuminated	Illuminated
2			Not illuminated

Note
- After the following times have elapsed, the operation step moves to the next step.
  - Step 1: **Approx. 5 s**
  - Step 2: **Approx. 25 s**
- When the last step is finished, the operation is repeated from Step 1.

Air intake actuator

Operation

Step	Air mix actuator	Airflow mode actuator	Blower speed	Magnet clutch	Air intake actuator	Rear window defroster
1	**0 %**	VENT	5th	ON	FRESH	OFF
2					RECIRCULATE	
3				OFF	FRESH	
4					RECIRCULATE	

Indicator light

Step	Recirculate switch indicator light	Rear window defroster switch indicator light	A/C switch indicator light
1	Not illuminated	Not illuminated	Illuminated
2	Illuminated		
3	Not illuminated		Not illuminated
4	Illuminated		

Note
- After **approx. 4 s** have elapsed, the operation step moves to the next step.
- When the last step is finished, the operation is repeated from Step 1.
- To prevent super-cooling, the magnetic clutch turns off despite the A/C forced override.

Blower motor

Operation

Step	Air mix actuator	Airflow mode actuator	Blower speed	Magnet clutch	Air intake actuator	Rear window defroster
1	**50 %**	VENT	OFF	OFF	FRESH	OFF
2			1st	ON		
3			3rd			
4			5th			
5			7th			

Indicator light

Step	Recirculate switch indicator light	Rear window defroster switch indicator light	A/C switch indicator light
1	Not illuminated	Not illuminated	Not illuminated
2			Illuminated
3			
4			
5			

Note
- After **approx. 4 s** have elapsed, the operation step moves to the next step.
- When the last step is finished, the operation is repeated from Step 1.
- To prevent super-cooling, the magnetic clutch turns off despite the A/C forced override.

Airflow mode actuator

Operation

Step	Air mix actuator	Airflow mode actuator	Blower speed	Magnet clutch	Air intake actuator	Rear window defroster
1	**50 %**	VENT	5th	ON	FRESH	OFF
2		BI-LEVEL				
3		HEAT				
4		DEF/HEAT				
5		DEFROSTER				

Indicator light

Step	Recirculate switch indicator light	Rear window defroster switch indicator light	A/C switch indicator light
1	Not illuminated	Not illuminated	Illuminated
2			
3			
4			
5			

Note

- After **approx. 4 s** have elapsed, the operation step moves to the next step.
- When the last step is finished, the operation is repeated from Step 1.
- To prevent super-cooling, the magnetic clutch turns off despite the A/C forced override.

Indicator light

Operation

Step	Air mix actuator	Airflow mode actuator	Blower speed	Magnet clutch	Air intake actuator	Rear window defroster
—	**50 %**	VENT	OFF	OFF	FRESH	OFF

Indicator light

Step	Recirculate switch indicator light	Rear window defroster switch indicator light	A/C switch indicator light
—	Illuminated	Illuminated	Illuminated

A/C operation check stop

Note

- If “Stop” is selected while the A/C operation check is operating, the A/C stops at the conditions shown in the table.

Operation

Step	Air mix actuator	Airflow mode actuator	Blower speed	Magnet clutch	Air intake actuator	Rear window defroster
—	**50 %**	VENT	OFF	OFF	FRESH	OFF

Indicator light

Step	Recirculate switch indicator light	Rear window defroster switch indicator light	A/C switch indicator light
—	Not illuminated	Not illuminated	Not illuminated